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scarlet2: Astronomical scene modeling in jax

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Summary

Large astronomical imaging surveys now contain billions of celestial sources, from stars in our Galaxy to galaxies towards the edge of the observable Universe. Because of improvements in resolution and sensitivity of current and future observing instruments, the images reveal more information than ever before. But they are also more difficult to analyze because galaxies exhibit complex morphologies, which cannot be described by traditional parametric models. And because there are so many sources, they routinely overlap with each other, either due to physical interactions or due to their close alignment along the line of sight. To extract all information of interest and avoid biases from incorrect modeling assumptions, it is therefore necessary to simultaneously model full scenes comprising many sources instead of analyzing each source separately, and each of the source models may itself need to be composed of multiple, morphologically complex components.

Statement of need

`scarlet2` is a Python package for full-scene modeling in observational astronomy. It inherits modeling assumptions from `scarlet` (Melchior et al., 2018), namely that a scene comprises multiple sources, each source comprises multiple components, and each component is determined by a spectrum model and a morphology model, whose outer product represents the light emission in a sky region as a hyperspectral data cube (wavelength $\times$ height $\times$ width). `scarlet2` retains the object-oriented paradigm and many classes and functions from `scarlet`, but augments standard Python with the `jax` library (Bradbury et al., 2018) and the `equinox` package (Kidger & Garcia, 2021) for automatic differentiation and just-in-time compilation.

`scarlet2` acts as a flexible, modular, and extendable modeling language for celestial sources that combines parametric and non-parametric models to describe complex scenarios such as multi-source blending, strong-lensing systems, supernovae and their host galaxies, etc. As a modeling language, `scarlet2` is agnostic about the optimization or inference method the user wants to employ, but it also provides methods to optimize the likelihood function or sample from the posterior, which utilize the `optax` package (DeepMind et al., 2020) or the `numpyro` inference framework (Bingham et al., 2019; Phan et al., 2019), respectively. The likelihood of multiple observations (at different resolutions, wavelengths, or observing epochs) can be combined for a joint model of static and transient sources. To match the coordinates from different observations, `scarlet2` utilizes the `Astropy` package (Astropy Collaboration, 2013). `scarlet2` can also interface with deep learning methods. Besides being natively portable to GPUs, parameters can be specified with neural networks as data-driven priors, which helps break the degeneracies that arise when multiple components are fit simultaneously (Sampson et al., 2024).

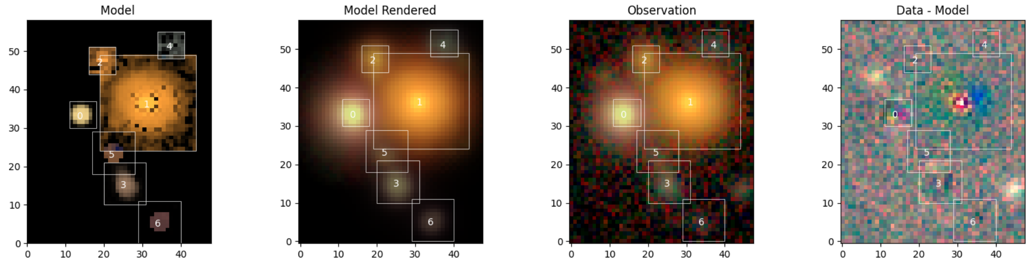

Figure 1: Scene with seven detected sources in multi-band images from the Hyper Suprime-Cam Subaru Strategic Program. Each source is modelled with a non-parametric spectrum and morphology (1st panel), the entire scene is then convolved with the telescope's point spread function (2nd panel) and compared to the observations (3rd panel). The residuals (4th panel) reveal the presence of previously undetected sources and source components (e.g. in the center of source #1).

To support the wide range of scientific studies that will be made with large sky surveys, `scarlet2` was designed with flexibility and ease of use in mind. Several publications have developed and demonstrated its capabilities, including modeling of interstellar dust embedded in distant galaxies (Siegel & Melchior, 2025) and of transient sources such as active galactic nuclei (Ward et al., 2025) and tidal disruption events (Yao et al., 2025). We have recently added support for users to make effective modeling choices and to validate their inference results. Future developments will integrate `scarlet2` into cloud-based science platforms and create a robust processing pipeline for joint pixel-level analyses of surveys from the Vera C. Rubin Observatory, the Euclid mission, the Nancy Grace Roman Space Telescope, and the La Silla Schmidt Southern Survey.

State of the field

`GALFIT` (Peng et al., 2002) implements parametric Sérsic model fitting in single-band, single-epoch data. The underlying algorithms are closed source and released only as precompiled binaries. `GALAPAGOS` (Barden et al., 2012) provides wrappers around `GAFLIT` in the proprietary IDL programming language; `GALAPAGOS-2` (Häußler et al., 2013) added multi-band fitting capabilities. `The Tractor` (Lang et al., 2016) is implemented in python and allows modeling galaxies in multi-band imaging at different resolutions as well as posterior sampling, but remains limited to parametric source profiles. `AstroPhot` (Stone et al., 2023) offers an astronomical modeling framework based on PyTorch (Paszke et al., 2019), which allows to optimize multiple sources with different parametric models to multiple images at different resolutions, and includes posterior sampling techniques. None of the implementations above support fully non-parametric galaxy models or neural network priors.

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In addition to dependencies mentioned earlier, `scarlet2` makes use of `numpy` (Harris et al., 2020), `matplotlib` (Hunter, 2007), and `h5py` (Collette, 2013).

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